BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: CPU, Memory and Storage

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. For a numeric age input, give erroneous data of the wrong type.

2. What does a record contain?
   A) Only one unnamed value
   B) Only network addresses
   C) Only one fixed data type
   D) Named fields about one item

3. "Correcting one line proves the entire program is now fault-free without further testing." Correct this. Explain why.

4. A program calculates an area and shows a menu in several places. Correct both claims. Explain why. (a) A function cannot return a value. (b) A procedure must return a value.

Today's objectives

Explain the core ideas of paper 1: cpu, memory and storage using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

CPU registers

PC, MAR, MDR and ACC have distinct jobs during instruction processing.

Primary memory

RAM is volatile working memory; ROM non-volatile firmware; cache accelerates access.

Storage selection

Compare SSD, HDD and optical media against contextual constraints.

Extended responses

Explain causal mechanisms, not merely 'faster' or 'cheaper'.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

CPU registers
PC, MAR, MDR and ACC have distinct jobs during instruction processing.

Primary memory
RAM is volatile working memory; ROM non-volatile firmware; cache accelerates access.

Storage selection
Compare SSD, HDD and optical media against contextual constraints.

Extended responses
Explain causal mechanisms, not merely 'faster' or 'cheaper'.

I Do - Worked model

Explain how cache can help a CPU running a frequently repeated program.

- Repeated instructions/data may be present in cache.
- Cache access is typically faster than RAM.
- CPU may avoid slower RAM fetches on a cache hit.
- This can improve execution performance.

We Do - Guided practice

1. A mobile field worker needs portable storage. Explain one benefit and drawback of SSD.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the final result after two assignments to a working variable.

```
acc = 6
acc = acc + 5
print(acc)
```

You Do - Independent exam questions

Q2. What does MDR hold during a memory operation? [2 marks]

Q3. Give one difference between RAM and ROM. [2 marks]

Q4. Explain one disadvantage of an HDD in a moving laptop. [3 marks]

Exit Ticket - True or False?

ROM is the volatile store for currently running applications.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The text 'twelve' is the wrong data type and must be handled or rejected.

2. D - Named fields about one item

3. To check that the fault is fixed and that other required behaviour still works.

4. (a) A function returns a value, such as an area. (b) A procedure performs a task, such as showing a menu, without returning a value. Use arguments and return values to avoid unwanted changes to global data.

I Do - Explain the model

Repeated instructions/data may be present in cache.

Cache access is typically faster than RAM.

CPU may avoid slower RAM fetches on a cache hit.

This can improve execution performance.

We Do - Model answers

1. Give scenario-linked balanced conclusion.

Code First - Answer

```
acc = 6
acc = acc + 5
print(acc)
```

Expected output: 11

You Do - Self-mark

Q2: Data or instruction being transferred.

Q3: RAM is volatile working memory; ROM is non-volatile and commonly stores firmware.

Q4: Moving mechanical parts make it more vulnerable to physical shock.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.